

2025 Ch H1 Q4

Section: Nature's Chemistry

Topic: Systematic Carbon Chemistry

Question summary

Identify the correct systematic name for the displayed alkene.

Worked solution

- 1) Choose the longest carbon chain that includes the C=C bond: taking ethyl–C=C–ethyl gives 6 carbons -> hex-.
- 2) Number the chain to give the lowest locant to the double bond -> hex-3-ene.
- 3) Substituents: each alkene carbon bears one extra CH₃ group -> methyls at C3 and C4.
- 4) Put substituents with multiplicative prefixes and positions:
3,4-dimethylhex-3-ene.

Final answer

D. 3,4-dimethylhex-3-ene

Revision tips

- Main chain must include the double bond and be as long as possible.
- Number from the end that gives the double bond the lowest possible number; then place substituent locants.
- List substituents alphabetically; use di-, tri- etc. but ignore these when alphabetising.